

**Real-World Problems and Significant Digits**Question Bank 4.04 **ANSWER KEY**

Name

Date

Period

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Show your work. Circle your final answer.

**For questions 1 through 4, give the number of significant digits in each measurement.**

1. 4,207 grams

**4**

*Every digit counts: the zero sits between two nonzero digits.*

2. 0.0058 meters

**2**

*Leading zeros are never significant.*

3. 3.200 seconds

**4**

*The trailing zeros count, because a decimal point is written. 3.200 and 3.2 are the same number but not the same claim.*

4. 52,000 litres

**2**

*No decimal point is written, so the trailing zeros are only holding places.*

**For questions 5 through 8, solve. Report each answer with the correct precision.**

5. A rectangular plot measures  $3.5 \times 10^2$  m by  $1.24 \times 10^2$  m. Find its area.

**$4.3 \times 10^4$  square meters**

*43,400 exactly; 2 significant digits and 3, so the answer keeps 2.*

6. A car travels  $2.60 \times 10^2$  km in 3.0 hours. Find its average speed.

**$8.7 \times 10^1$  km/h**

*$260 \div 3.0 = 86.66\dots$ ; 3 significant digits and 2, so the answer keeps 2.*

7. Two lengths are measured:  $8.7 \times 10^2$  mm and  $4.5 \times 10^2$  mm. Find their sum.

**$1.32 \times 10^3$  mm**

*870 and 450 both stop at the tens place, so the sum, 1,320, does too. A sum can carry more significant digits than either measurement — three here, from two twos.*

8. Two masses are measured:  $8.40 \times 10^{-2}$  g and  $3.715 \times 10^{-3}$  g. Find their difference.

$8.03 \times 10^{-2}$  g

*0.080285 exactly; 0.0840 stops at the ten-thousandths place and 0.003715 stops two places further out, so the answer stops at the ten-thousandths.*

### Real quantities.

9. The Amazon River discharges about  $5.5 \times 10^7$  gallons of water each second.

- a. How many gallons does it discharge in one hour, which is  $3.6 \times 10^3$  seconds? Report the answer with the correct number of significant digits.

$2.0 \times 10^{11}$  gallons

*$1.98 \times 10^{11}$  exactly. The rate carries two significant digits; 3,600 seconds in an hour is an exact conversion, not a measurement, so it does not limit the count. The answer keeps two.*

- b. A second river discharges about  $1.10 \times 10^7$  gallons per second. How many times larger is the Amazon's rate? Report the answer with the correct number of significant digits.

**5.0 times larger**

*Exactly 5; two significant digits are justified, so it is written 5.0.*

10. A processor performs about  $4.66 \times 10^8$  calculations each second.

- a. How many calculations does it perform in one minute? Report the answer with the correct number of significant digits.

$2.80 \times 10^{10}$  calculations

*$4.66 \times 10^8 \cdot 60 = 2.796 \times 10^{10}$ . Sixty seconds in a minute is exact, not a measurement, so it does not limit the count — the answer keeps the measurement's three digits.*

- b. Five of these processors run for 20 minutes. How many calculations do they perform altogether? Report the answer with the correct number of significant digits.

$2.80 \times 10^{12}$  calculations

*$2.796 \times 10^{10} \cdot 5 \cdot 20 = 2.796 \times 10^{12}$ . Five and twenty are counts, so again only the measurement's three digits survive.*

### Reading someone else's work.

11. Omar multiplies a measured length of  $2.5 \times 10^2$  m by a measured width of  $1.346 \times 10^2$  m and reports the area as  $3.365 \times 10^4$  square meters. Is his arithmetic correct? Is his answer correct? Explain.

**His arithmetic is correct — the exact product is  $3.365 \times 10^4$ . His answer is not, because 2.5 carries only two significant digits, so the area can carry only two:  $3.4 \times 10^4$  square meters.**

*A correct value in the wrong form is still the wrong answer here; the rule is part of the question.*

12. A rectangle's sides are measured as  $7.2 \times 10^1$  cm and  $4.0 \times 10^1$  cm. Which is the area, in scientific notation with the correct number of significant digits?

- ☐ A.  $2.9 \times 10^2$  cm<sup>2</sup>
- ☐ B.  $28.8 \times 10^2$  cm<sup>2</sup>
- ☒ C.  $2.9 \times 10^3$  cm<sup>2</sup>
- ☐ D.  $2.88 \times 10^3$  cm<sup>2</sup>

C